Autrin Hakimi

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PROFESSIONAL PROFILE

Software Systems Engineer specializing in **full-stack applications**, **AI/ML systems**, and **robotics automation**. Leads architecture, development, and deployment of production systems for semiconductor industry, delivering software for a **seven-figure robotics** product and a high-impact ML-driven parts-matching tool. Published researcher with **ACM SIGSPATIAL 2025** paper on **distributed deep learning** across **64 NVIDIA A100 GPUs**. Seeking to apply full-stack and AI expertise to drive innovative, high-value solutions in a forward-thinking engineering team.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Bash, HTML, CSS, C#
Web & Backend: React, Python Flask, Tkinter, REST APIs, WebSockets
Cloud & DevOps: Docker, Git, CI/CD pipelines, PyInstaller/NSIS packaging, Linux/Unix, Microsoft Azure, Google Cloud
ML & Data: PyTorch, scikit-learn, Pandas, NumPy, distributed training, DGL, NetworkX
Databases: MySQL, SQL, SQLite, NoSQL, Snowflake
Automation & Control: Robot workflow orchestration, vision systems, HMI/GUI, ROS, OpenCV, robot controllers, automated testing, multithreading
Methodologies: Agile (Scrum/Kanban), test-driven development, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Software Systems Engineer
Jan 2026 - Present
ASM International N.V. — *Phoenix, AZ*

- **Own architecture and delivery** of multiple production applications spanning **full-stack application development**, **robotics**, **computer vision**, and **machine learning (ML)-based parts matching**; **independently** define technical requirements, **make architecture decisions**, and deploy and support releases at international manufacturing sites.
- Authored most of the **backend and frontend** for an operator application shipping with a **seven-figure** collaborative robotics product; built the **human-machine interface** (**HMI)**, **robot controls**, **responsive camera integration**, **thread** synchronization, communications recovery, and safe abort and return-to-home sequences. The project received a quarterly team **award**.
- Improved measurement accuracy and reliability on a **robot-and-camera** conductance fixture by correcting motion, calibration, and geometry logic across part configurations; built 140+ automated tests to validate geometry and control behavior without physical hardware.
- Built a full-stack ML parts-matching application for **chamber optimization** and **chamber/parts health prediction**; the system forms the basis of a pending patent on which I am a named inventor.
- **Mentor senior and principal engineers**, **conduct technical interviews** for engineering candidates including senior roles, and **coordinate implementation** across teams; proactively communicate progress and risks to management. Introduced Git version control to improve code traceability.

Undergraduate AI Researcher
Aug 2024 - Nov 2025
Iowa State University, SwAPP Lab — *Remote*

- Co-authored 'HydroGAT' accepted at **ACM SIGSPATIAL 2025**: a **heterogeneous graph attention transformer** for flood prediction reaching NSE **0.97**; built **distributed training** across **64 NVIDIA A100 GPUs** on NERSC Perlmutter with **15× speedup**
- Engineered end-to-end data pipelines (**PyTorch, DGL, Pandas, NumPy**) for large-scale datasets; constructed **heterogeneous graphs**, built **AI models**, automated workflows via **Bash** scripting, cutting setup time **40%**
- Code: https://github.com/swapp-lab/HydroGAT Paper: https://dl.acm.org/doi/10.1145/3748636.3764172

Software Development Intern
May 2024 - Aug 2024
BuilderTREND — *Omaha, NE*

- Designed and shipped a React-based job proposal template, migrating legacy ASP pages to React and building full-stack features with JavaScript, TypeScript, and C# (.NET) during Kanban/Scrum sprints, which streamlined proposal creation and improved user experience

Data Science Intern
May 2023 - May 2024
Alliant Energy — *Remote*

- Developed predictive **ML models** (**Random Forest, SVM**) in **Python** and **SQL** analyzing electricity usage for **995K** customers in Snowflake; predicted EV ownership for customer segmentation, recognized by the Director of Data Analytics for impactful business insights.

PROJECTS

Defect Sense code

- Built a local industrial visual-inspection system using **PatchCore/Anomalib** and **FastAPI**, with **anomaly heatmaps**, **region localization**, and optional schema-constrained **VLM** reports.
- Evaluated on 63 MVTec AD bottle images with separate training and calibration sets: detected **48/48** defects with **0/15** false alarms, averaging **0.30 s/image** on an RTX 4070 Laptop GPU; added 145 GPU-independent tests.

Autonomous Robotic Chess Player code

- Built a fully **autonomous** system: piece recognition (**OpenCV**), **move calculation**, and **UR10e robot** arm control: **ROS, MoveIt**

EDUCATION

B.S., Computer Science | *Iowa State University* | Aug 2021 - May 2025
GPA: 3.70/4.0 | Magna Cum Laude